

Santhosh Ganesan

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Summary

AI and Software Engineering student with hands-on experience building intelligent systems using Machine Learning, Computer Vision, NLP, and Edge AI technologies. Skilled in developing real-world AI pipelines, optimizing models for edge deployment, and building backend-integrated ML applications using FastAPI, Streamlit, and ONNX.

Skills

Programming Languages: Python, Java, C/C++, JavaScript, HTML, CSS

Machine Learning & AI: PyTorch, Scikit-Learn, NLP, SBERT, GNNs, YOLOv5, NumPy, Matplotlib

Backend & Deployment: FastAPI, Flask, Streamlit, Docker, ONNX

IoT & Edge AI: Arduino, ESP32, Raspberry Pi, Edge Impulse

Core CS Concepts: Data Structures & Algorithms, OOP, Computer Networks, Embedded Systems

Tools & Platforms: Git, Linux, Windows

Experience

AI Intern

May 2025 – Jul 2025

Innominds Software Pvt. Ltd, Bengaluru

- Built a YOLOv5-based detection pipeline for identifying safety helmets and vests, achieving 88% detection accuracy across validation samples.
- Converted and optimized multiple trained models into ONNX format to reduce inference overhead for edge deployment environments.
- Integrated Flask-based API endpoints for serving computer vision inference results to external applications.
- Applied custom DSP blocks within Edge Impulse pipelines to support deployment on resource-constrained IoT hardware.
- Assisted in debugging model inference workflows and preparing technical documentation for deployment testing.

Projects

Predictive Modelling of Chip Routability via Graph Neural Networks (GNNs)

PyTorch Geometric, GNNs, Python

- Modelled more than 170 IC layouts as grid-based graph structures and trained a GCN-based Graph Neural Network for routability classification.
- Generated pseudo-labels using congestion and DRC threshold analysis to enable supervised learning on unlabeled layout datasets.
- Produced routability heatmaps through a custom inference pipeline, enabling visual separation between congested and optimized layouts.

AI Predictive Maintenance System

Machine Learning, FastAPI, Streamlit, Industrial IoT

- Processed multivariate sensor streams using Isolation Forest and regression models to estimate Remaining Useful Life (RUL).
- Engineered drift-resistant features to improve anomaly detection robustness across varying industrial machine conditions.
- Built a FastAPI backend and Streamlit dashboard for real-time monitoring, prediction serving, and machine health visualization.

AI-Powered Resume Analysis & Job Recommendation System

NLP, SBERT, OCR, Streamlit

- Parsed and processed resume documents using OCR-based preprocessing and a modular NLP extraction pipeline.
- Computed semantic similarity scores using SBERT embeddings and cosine similarity, achieving 83% validation accuracy for candidate-role matching.
- Created an interactive Streamlit dashboard for resume analytics, profile scoring, and AI-driven job recommendations.

Certifications

- Getting Started with Deep Learning – NVIDIA
- Edge AI – Edge Impulse
- Supervised Machine Learning – Coursera

Education

B.Tech. in Computer Science

Aug 2022 – Jul 2026

Vellore Institute of Technology, Vellore | *GPA: 8.03 / 10*

Senior Secondary Education

Aug 2020 – May 2022

Sri Sri Ravishankar Vidya Mandir, Bengaluru | *Percentage: 89.2%*